FOUNDATION COURSE

START YOUR PREPARATION FROM BASICS

Prerequisites of Engineering Mathematics

Lecture no. 10

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Recap *of previous lecture*

VECTORS

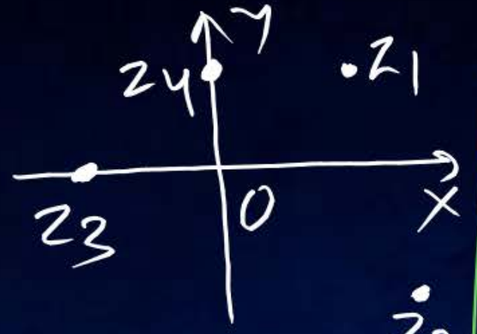


Topics *to be covered*

"COMPLEX"



Complex Numbers



$z_1 = 2+3i \simeq (2,3)$

$z_2 = 4-5i \simeq (4,-5)$

$z_3 = -3 \simeq -3+0.i \simeq (-3,0)$

$z_4 = 4i = 0+4i \simeq (0,4)$

⊛ All the Real Nos are also Comp. Numbers in which Img part is zero (T)

⊛ If Real Part is zero then it is Purely Img No.
of Img " " " " Purely Real No.

⊛ Def^n :-

$\boxed{z = x+iy} \simeq (x,y)$ where → Real Part $= x$, Img part $= y$

But Both the parts are Real Numbers.

⊛ Conjugate of Complex No →

By Replacing i with $-i$, we can get Conj of any Comp. Number. i.e

$z = x+iy$

$\bar{z} = x-iy$

$\frac{z+\bar{z}}{2} = x$ is Real Part

$\frac{z-\bar{z}}{2i} = y$ is Img Part.

(*) Consider the eqn $z^2+1=0$

$z^2=-1 \Rightarrow z=\pm\sqrt{-1}$

$z=\pm i$

Let $\sqrt{-1}=i$

(*) $\boxed{i=\sqrt{-1}}$

$i^2=-1$

$i^3=i^2 i=-i$

$i^4=i^2 i^2=(-1)(-1)=1$

$i^0=1$ (let)

$i^{-1}=\frac{1}{i}=\frac{i}{i^2}=-i$

$i^{-2}=\frac{1}{i^2}=-1$

$i^{-3}=\frac{1}{i^3}=\frac{i}{i^4}=i$

$i^{-4}=\frac{1}{i^4}=1$

ie $\boxed{i=\sqrt{-1},\ i^2=-1,\ i^3=-i,\ i^4=1,\ i^0=1}$

& $\boxed{i^{-1}=-i,\ i^{-2}=-1,\ i^{-3}=i,\ i^{-4}=1}$

(*) $\boxed{i^n=i^r}$ where r = Remainder when $\left(\frac{n}{4}\right)$

eg $i^{18}=?$

(M-I) $i^{18}=i^{16}\cdot i^2=(i^4)^4\cdot i^2=1\cdot i^2=i^2=-1$

(M-II) Rem when $\left(\frac{18}{4}\right)=2$ So $i^{18}=i^2=-1$

eg $i^{43}=?=i^3=-i$

$\because$ Rem when $\left(\frac{43}{4}\right)=3$

eg: $i^{48}=?=i^0=1$

PQ: $i^0+i^2+i^4+i^6+i^8+\ldots\ldots+i^{20}=?$

W.K.T that $\boxed{i^n=i^r}$ where $r=$ Rem when $\left(\frac{n}{4}\right)$

$i^0+i^2+i^4+i^2+i^0+i^2+i^0+i^2+i^0$

$+i^2+i^0$

$=1-1+1-1+1-1+1-1+1-1+1=1$

Q Evaluate $i^{63}+i^{-64}+i^{-65}+i^{66}=?$

$=i^3+i^0+i^1+i^2=-i+1+i-1=0$

(*) $\boxed{i^n+i^{n+1}+i^{n+2}+i^{n+3}=0}$

Q The conjugate of $\frac{1}{3+4i}$ will be?

Sol: $z=\frac{1}{3+4i}\Rightarrow \bar{z}=\frac{1}{3-4i}$ Ans.

i.e. $\bar{z}=\frac{1}{3-4i}\times\left(\frac{3+4i}{3+4i}\right)=\frac{3+4i}{3^2-(4i)^2}$

$(a+b)(a-b)=a^2-b^2$

$=\frac{3+4i}{9-16i^2}=\frac{3+4i}{9-16(-1)}=\frac{3+4i}{25}$

$\bar{z}=\frac{3}{25}+\left(\frac{4}{25}\right)i$ Ans.

(*) The Conjugate of $\frac{1+i}{1-i}$ will be ?

Sol: $z = \frac{1+i}{1-i} = \frac{1+i}{1-i} \times \left(\frac{1+i}{1+i}\right) = \frac{(1+i)^2}{1^2 - i^2}$

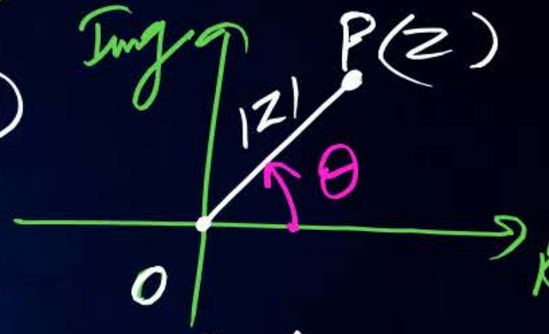
$= \frac{1^2 + i^2 + 2(1)(i)}{1-(-1)} = \frac{1-1+2i}{2}$

$z = \boxed{\frac{1+i}{1-i} = i}$ Learn too $\bar{z} = -i$

(*) The Value of $\left(\frac{1+i}{1-i}\right)^{13} = ? = (i)^{13} = i^1 = i$

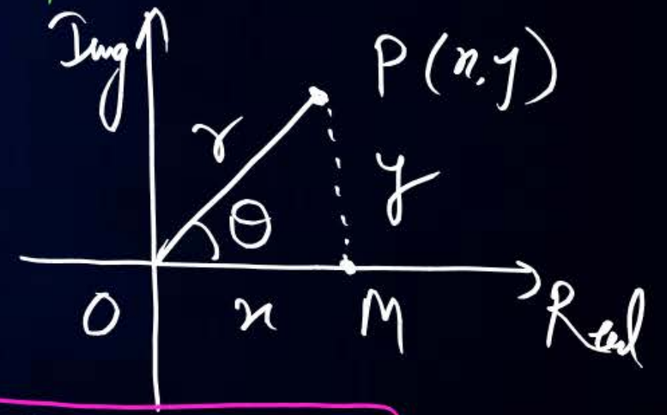
Modulus and Argument of Complex No :-

$\boxed{z = x+iy} = (x, y)$



$|z|$ = Modulus of z

θ = Argument of z



$OP^2 = OM^2 + MP^2$

$r^2 = x^2 + y^2$

$|z|^2 = x^2 + y^2 \Rightarrow \boxed{|z| = \sqrt{x^2+y^2}}$

$\tan\theta = \frac{y}{x} \Rightarrow \theta = \tan^{-1}\left(\frac{y}{x}\right)$

i.e. for $z = x+iy$ $\Rightarrow$ $r = |z| = \sqrt{x^2+y^2}$

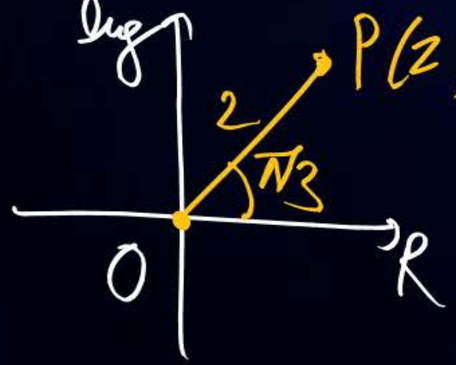
$\theta = \text{Arg}(z) = \tan^{-1}\left(\frac{y}{x}\right)$

Q: Find the Modulus and Arg of $1+i\sqrt{3}$

Sol: $z = 1+i\sqrt{3}$ (Ist Quad.)

$|z| = \sqrt{1^2+(\sqrt{3})^2} = \sqrt{1+4} = 2$

$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{\sqrt{3}}{1}\right) = \frac{\pi}{3}$



Q: Find the Modulus and Arg of $-1-i\sqrt{3}$ (III Quad.)

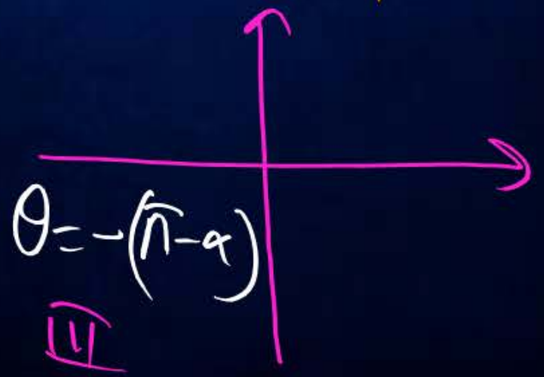
Sol: $z = -1-i\sqrt{3} \Rightarrow |z| = \sqrt{(-1)^2+(-\sqrt{3})^2} = 2$

~~$\text{Arg}\, z = \theta = \tan^{-1}\left(\frac{-\sqrt{3}}{-1}\right) = \tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$~~

this approach is not good enough.

Correct Approach $z = -1-i\sqrt{3}$ (III) $\Rightarrow$ $x = -1$, $y = -\sqrt{3}$

$\alpha = \tan^{-1}\left|\frac{y}{x}\right| = \tan^{-1}\left|\frac{-\sqrt{3}}{-1}\right| = \tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$



So $\theta = -\left(\pi-\frac{\pi}{3}\right) = -\frac{2\pi}{3}$

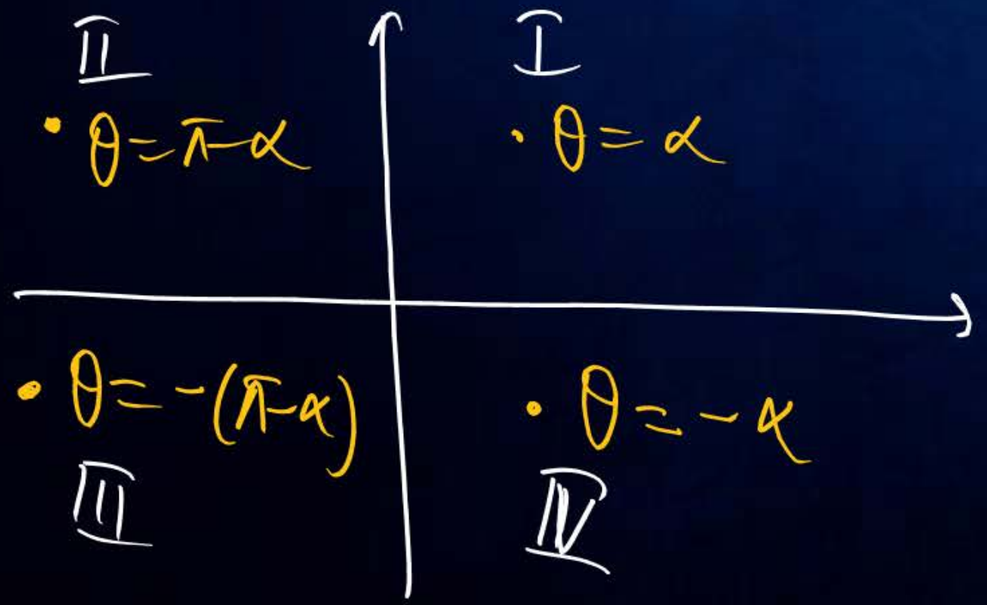
Principle Values of Arguements :-

Those values of θ which lies b/w $-\pi$ & π are called P. Values i.e $-\pi < \theta \leq \pi$

Method of finding P. Values of θ → Let $z = x + iy$

then first find α using $\alpha = \tan^{-1}\left|\frac{y}{x}\right|$.

then θ will be as follows.



Q: Find Mod & Arg of $(\pm 1 \pm i\sqrt{3})$

Sol:

$z_1 = 1 + i\sqrt{3}$ (I)

$z_2 = 1 - i\sqrt{3}$ (IV)

$z_3 = -1 + i\sqrt{3}$ (II)

$z_4 = -1 - i\sqrt{3}$ (III)

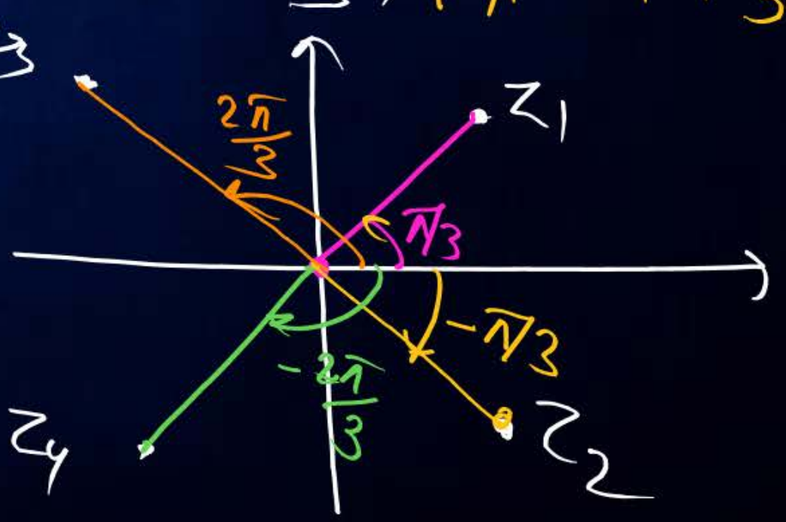
$\Rightarrow \theta_1 = \pi/3$

$\theta_2 = -\pi/3$

$\theta_3 = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$

$\theta_4 = -\left(\pi - \frac{\pi}{3}\right) = \frac{-2\pi}{3}$

$\alpha = \tan^{-1}\left|\frac{y}{x}\right| = \frac{\pi}{3}$, $|z_1| = |z_2| = |z_3| = |z_4| = 2$



Euler Notation →

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots\cdots$$

i.e. $$e^z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \frac{z^4}{4!} + \cdots\cdots$$

Put $z = i\theta$

$$e^{i\theta} = 1 + (i\theta) + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \frac{(i\theta)^4}{4!} + \cdots\cdots$$

$$= \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \cdots\right) + i\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \cdots\right)$$

$$e^{i\theta} = \cos\theta + i\sin\theta$$

i.e. $\boxed{e^{i\theta} = \cos\theta + i\sin\theta}$

where $|e^{i\theta}| = \sqrt{\cos^2\theta + \sin^2\theta} = 1$

(*) $|i| = ?$

Let $z = i = 0 + (1)i$

$$|z| = |i| = \sqrt{0^2 + 1^2} = 1$$

i.e. $\boxed{|i| = 1}$ Similarly $|-i| = 1$.

(*) Conclusions : $\boxed{e^{i\theta} = \cos\theta + i\sin\theta}$, $|e^{i\theta}| = 1$

eg $e^{2\pi i} = ? = \cos(2\pi) + i\sin(2\pi) = 1 + i(0) = 1$

eg $e^{\pi i} = ? = \cos(\pi) + i\sin(\pi) = -1 + i(0) = -1$

eg $e^{\frac{\pi}{2} i} = ? = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2} = 0 + i(1) = i$

eg $e^{-\frac{\pi}{2} i} = ? = \cos\left(-\frac{\pi}{2}\right) + i\sin\left(-\frac{\pi}{2}\right)$

$= \cos\frac{\pi}{2} - i\sin\frac{\pi}{2} = 0 - i(1) = -i$

ie $\boxed{1 = e^{2\pi i},\ -1 = e^{\pi i},\ i = e^{\frac{\pi}{2} i},\ -i = e^{-\frac{\pi}{2} i}}$

eg the imaginary part of $\log_e(\sqrt{i}) = ?$

Sol: $z = \log\sqrt{i} = \log(i)^{\frac{1}{2}} = \frac{1}{2}\log(i)$

$= \frac{1}{2}\log_e\left(e^{\frac{\pi}{2} i}\right) = \frac{1}{2}\left(\frac{\pi}{2} i\right)\log_e e$

$z = \frac{\pi}{4} i = 0 + \left(\frac{\pi}{4}\right) i$

$\therefore \text{Img}(z) = \pi/4$.

eg: Evaluate $i^i = ?$

$i^i = \left(e^{\frac{\pi}{2} i}\right)^i = e^{\frac{\pi}{2} i^2} = e^{-\frac{\pi}{2}}$ Ans

✳ $e^{i\theta} = \cos\theta + i\sin\theta$

& $e^{-i\theta} = \cos\theta - i\sin\theta$

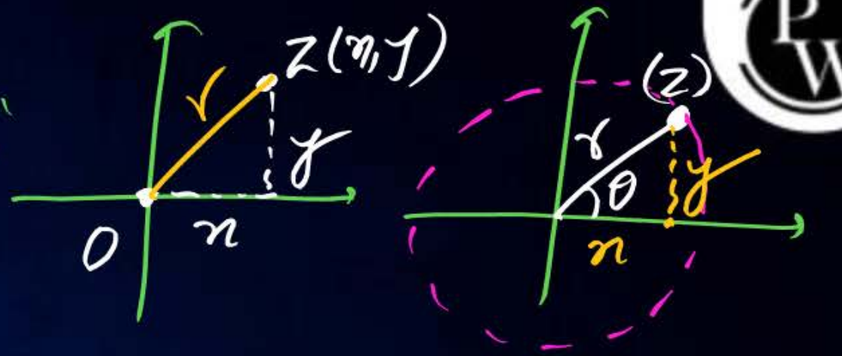
$\frac{e^{i\theta}+e^{-i\theta}}{2} = \cos\theta$

$\frac{e^{i\theta}-e^{-i\theta}}{2i} = \sin\theta$

eg Evaluate $e^{\frac{5\pi}{2}i} - e^{-\frac{5\pi}{2}i} = ? = 2i\sin\left(\frac{5\pi}{2}\right)$

$= 2i\sin\left[2\pi + \frac{\pi}{2}\right] = 2i\sin\frac{\pi}{2} = 2i$

Note:



$\frac{x}{r} = \cos\theta$, $\frac{y}{r} = \sin\theta$

$\Rightarrow x = r\cos\theta$, $y = r\sin\theta$ (Polar Coord.)

✳ Representation of Comp Numbers → Let $z = x+iy$

$$z = x+iy = r(\cos\theta + i\sin\theta) = r.e^{i\theta}$$

C. form Polar form E. form

where $r = |z| = \sqrt{x^2+y^2}$

$\theta = \text{Arg}(z) = \tan^{-1}\left(\frac{y}{x}\right)$

eg Evaluate $(1+i\sqrt{3})^{24} = ?$

Sol: Let $z = 1+i\sqrt{3}$

$r = |z| = 2$

$\theta = \text{Arg } z = \frac{\pi}{3}$

$\therefore (1+i\sqrt{3})^{24} = z^{24} = (re^{i\theta})^{24}$

$= \left(2.e^{i\frac{\pi}{3}}\right)^{24} = 2^{24}.e^{8\pi i}$

$= 2^{24}\left(e^{2\pi i}\right)^4 = 2^{24}.(1)^4 = 2^{24}$ //

GATE

eg the value of $(1+i)^8 = ?$

Let $z = 1+i$

$r = |z| = \sqrt{1^2+1^2} = \sqrt{2}$

$\theta = \text{Arg } z = \tan^{-1}\left(\frac{1}{1}\right) = \frac{\pi}{4}$

$\therefore (1+i)^8 = z^8 = (r.e^{i\theta})^8$

$= \left(\sqrt{2}e^{i.\frac{\pi}{4}}\right)^8 = (\sqrt{2})^8 e^{2\pi i}$

$= 2^4.(1) = 16$ //

De-Moivre's Theorem :- If $n \in I$ or $\mathbb{Q}$

$$\boxed{(\cos\theta \pm i\sin\theta)^n = \cos(n\theta) \pm i\sin(n\theta)}$$

$$\text{LHS} = (\cos\theta + i\sin\theta)^n = (e^{i\theta})^n$$

$$= e^{i(n\theta)} = \cos(n\theta) + i\sin(n\theta)$$

$$= \text{RHS}$$

Note:

1. $(\cos\phi + i\sin\theta)^n \neq \cos n\phi + i\sin n\theta$
2. $(\sin\theta + i\cos\theta)^n \neq \sin n\theta + i\cos n\theta$

Roots of Complex Number :-

Consider $z^{\frac{1}{n}}$ then it has n roots that can be obtained at $n = 0, 1, 2, 3, \ldots, (n-1)$

Cube roots of unity :-

$(1^{1/3}) \Rightarrow 1, \omega, \omega^2$ where $\omega = e^{\frac{2\pi}{3}i} = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$

Cube roots of (-1) :-

$(-1)^{\frac{1}{3}} \Rightarrow -1, -\omega, -\omega^2$ where $\omega = e^{\frac{2\pi}{3}i} = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$

Cube roots of unity → Consider $z^3 - 1 = 0 \Rightarrow z = (1)^{\frac{1}{3}}$

or $z = (e^{2\pi i})^{\frac{1}{3}} = (\cos 2\pi + i \sin 2\pi)^{\frac{1}{3}}$

$= [\cos(2n\pi + 2\pi) + i \sin(2n\pi + 2\pi)]^{\frac{1}{3}}$

$z = \cos\left(\frac{2n\pi + 2\pi}{3}\right) + i \sin\left(\frac{2n\pi + 2\pi}{3}\right)$ — (1)

for $n = 0$, $z_1 = \cos\left(\frac{2\pi}{3}\right) + i \sin\left(\frac{2\pi}{3}\right) = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$

for $n = 1$, $z_2 = \cos\left(\frac{4\pi}{3}\right) + i \sin\left(\frac{4\pi}{3}\right) = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$

for $n = 2$, $z_3 = \cos(2\pi) + i \sin(2\pi) = 1$

i.e Cube root of unity are as follows;

$(1)^{\frac{1}{3}} \rightarrow 1$

$\rightarrow -\frac{1}{2} + i\frac{\sqrt{3}}{2} = e^{\left(\frac{2\pi}{3}\right)i} = \omega$ (let)

$\rightarrow -\frac{1}{2} - i\frac{\sqrt{3}}{2} = e^{\left(\frac{4\pi}{3}\right)i} = \omega^2$

(1) $\because \omega^2 = \omega \cdot \omega = e^{\frac{2\pi}{3}i} \cdot e^{\frac{2\pi}{3}i} = e^{\frac{4\pi}{3}i}$

(2) $\bar{\omega} = \omega^2$, $\bar{\omega^2} = \omega$, $\frac{1}{\omega} = \frac{\omega^2}{\omega^3} = \omega^2$

(3) $\omega^3 = 1$, (4) $1 + \omega + \omega^2 = 0$

Cube root of (-1) → Consider $z^3+1=0$

$$\text{i.e } z=(-1)^{\frac{1}{3}}=(e^{\pi i})^{\frac{1}{3}}=(\cos\pi+i\sin\pi)^{\frac{1}{3}}$$

$$=\left[\cos(2n\pi+\pi)+i\sin(2n\pi+\pi)\right]^{\frac{1}{3}}$$

$$z=\cos\left(\frac{2n\pi+\pi}{3}\right)+i\sin\left(\frac{2n\pi+\pi}{3}\right) \quad \text{---(1)}$$

Take $n=0$, $z_1=\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}=\frac{1}{2}+i\frac{\sqrt{3}}{2}=-w^2$

Take $n=1$, $z_2=\cos\pi+i\sin\pi=-1$

Take $n=2$, $z_3=\cos\frac{5\pi}{3}+i\sin\frac{5\pi}{3}=\frac{1}{2}-i\frac{\sqrt{3}}{2}=-w$

Sq. roots of i → Consider $z^2-i=0$

$$z=(i)^{\frac{1}{2}}=\left(e^{\frac{\pi}{2}i}\right)^{\frac{1}{2}}=\left(\cos\frac{\pi}{2}+i\sin\frac{\pi}{2}\right)^{\frac{1}{2}}$$

$$=\left[\cos\left(2n\pi+\frac{\pi}{2}\right)+i\sin\left(2n\pi+\frac{\pi}{2}\right)\right]^{\frac{1}{2}}$$

$$z=\cos\left(n\pi+\frac{\pi}{4}\right)+i\sin\left(n\pi+\frac{\pi}{4}\right) \quad \text{---(1)}$$

For $n=0$, $z_1=\cos\frac{\pi}{4}+i\sin\frac{\pi}{4}=\frac{1}{\sqrt{2}}+i\left(\frac{1}{\sqrt{2}}\right)$

For $n=1$, $z_2=\cos\left(\frac{5\pi}{4}\right)+i\sin\left(\frac{5\pi}{4}\right)$

$$=-\frac{1}{\sqrt{2}}+i\left(-\frac{1}{\sqrt{2}}\right)$$

$$\text{i.e } (i)^{\frac{1}{2}}=\pm\frac{1}{\sqrt{2}}(1+i)$$

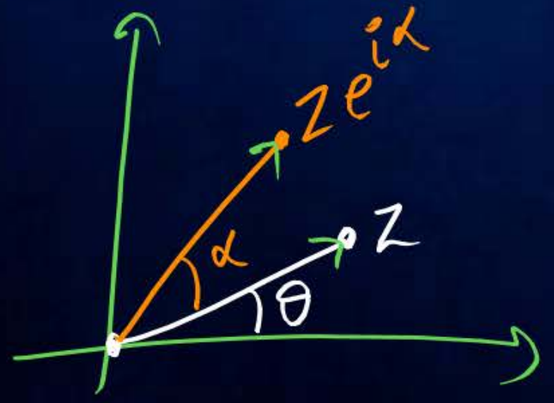
	Sq. roots	Cube roots
1	± 1	$1, \omega, \omega^2$
-1	$\pm i$	$-1, -\omega, -\omega^2$
i	$\pm \frac{1}{\sqrt{2}}(1+i)$	$-i, \pm \frac{\sqrt{3}}{2} + \frac{i}{2}$

Complex No as Rotating arrow →

Let $z = re^{i\theta}$

then $ze^{i\alpha} = re^{i\theta} \cdot e^{i\alpha}$

$= re^{i(\theta+\alpha)}$



i.e when we multiply z with $e^{i\alpha}$ it will rotate by angle α in an A.C.W sense

Similarly, if we multiply z with $e^{-i\alpha}$ then z will Rotate by angle α in a C.W sense

$\because ze^{-i\alpha} = re^{i(\theta-\alpha)}$

Eg Let $z = re^{i\theta}$ then angle b/w z & iz ?

Sol: $iz = e^{\frac{\pi}{2}i}\left(re^{i\theta}\right) = re^{i\left(\theta+\frac{\pi}{2}\right)}$

$\text{Arg}(z) = \theta$ & $\text{Arg}(iz) = \theta + \pi/2$

$\therefore$ Req Angle $= \pi/2$

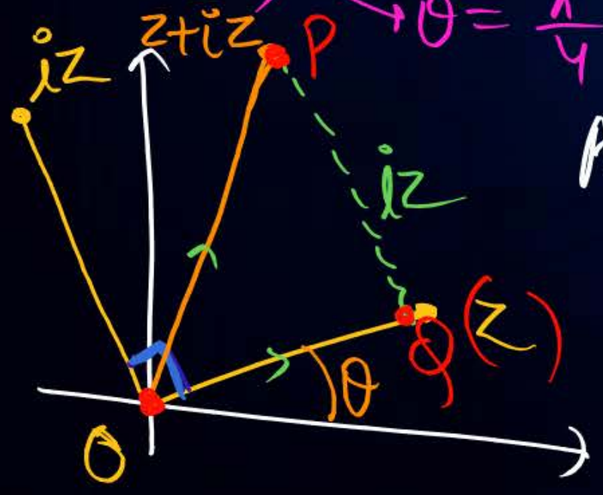
Q.e.

Find the area of the △ formed by z, iz & $z+iz$ where z is any Comp No.

Sol: Let $z = re^{i\theta}$, $iz = re^{i(\theta+\frac{\pi}{2})}$

$z+iz = z(1+i) = re^{i\theta} \cdot \sqrt{2}e^{i\frac{\pi}{4}} = (r\sqrt{2})e^{i(\theta+\frac{\pi}{4})}$

$\because (1+i) \Rightarrow r = \sqrt{2}, \theta = \frac{\pi}{4} \Rightarrow (1+i) = \sqrt{2}e^{i\frac{\pi}{4}}$



Area of $OPQ = \frac{1}{2} \times Base \times Ht$

$= \frac{1}{2} \times OQ \times QP$

$= \frac{1}{2}|z| \times |z| = \frac{|z|^2}{2}$

where $QP = |iz| = |i||z| = |z|$

Note: $w = e^{i(\frac{2\pi}{3})} = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$

w.k.that $|e^{i\theta}| = 1 \Rightarrow |w| = 1$

Similarly, $w^2 = e^{i(\frac{4\pi}{3})} = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$

$\Rightarrow |w^2| = 1$

& $1 + w + w^2 = 0 \Rightarrow 1 + w = -w^2$

Q: Find the area of the $\triangle$ formed by $z, wz, z+wz$ where $z = 3e^{i\theta}$

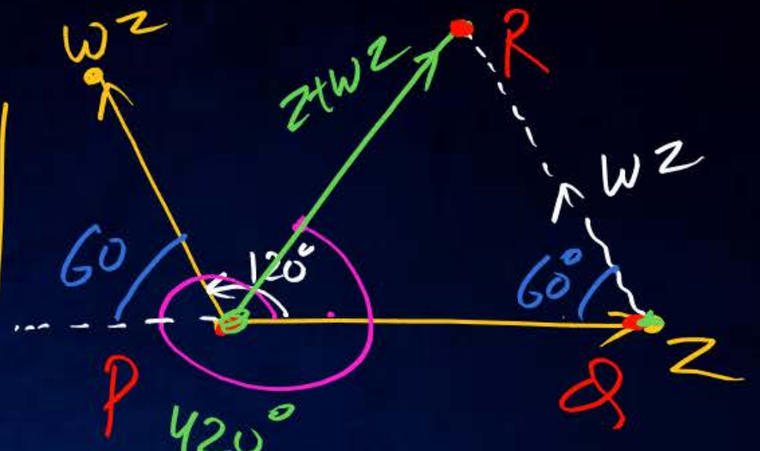
Sol: $z = 3e^{i\theta} \Rightarrow |z| = |3e^{i\theta}| = 3$

$wz = e^{\left(\frac{2\pi}{3}\right)i} \cdot 3e^{i\theta} = 3e^{i\left(\theta + \frac{2\pi}{3}\right)}$

& $z+wz = (1+w)z = -w^2 z$

$= \left(e^{\pi i}\right) e^{\left(\frac{4\pi}{3}\right)i} \cdot 3e^{i\theta}$

$= 3e^{\left(\theta + \frac{7\pi}{3}\right)i}$



$|wz| = |w||z| = |z|$

$|z+wz| = |(1+w)z| = |-w^2 z| = |w^2||z| = |z|$

i.e. $|z| = |wz| = |z+wz|$ so PQR is an equilateral $\triangle$

So $\text{Ar}(\triangle PQR) = \frac{\sqrt{3}}{4} \times (\text{side})^2 = \frac{\sqrt{3}}{4}|z|^2 = \frac{9\sqrt{3}}{4}$

Equⁿ of Circle in Complex Form →

(1) it is given as ;

$|z-z_0|=\gamma$ where → Centre $= z_0 = (x_0, y_0)$, rad $= \gamma$

Let $z = x+iy \simeq (x, y)$

$z_0 = x_0+iy_0 \simeq (x_0, y_0)$

$|z-z_0|^2 = \gamma^2$

$|(x-x_0)+i(y-y_0)|^2 = \gamma^2$

$(x-x_0)^2+(y-y_0)^2 = \gamma^2$ //

Note: $|z|^2 = z\bar{z} = x^2+y^2$

Particular Case: $|z|=\gamma$ → Centre $= 0 \simeq (0,0)$, rad $= \gamma$

eg $|z|=5$ circle → Centre $=(0,0)$, rad $=5$

(2) $|z-z_0|=\gamma \Rightarrow |z-z_0|^2 = \gamma^2$

$(z-z_0)\overline{(z-z_0)} = \gamma^2$

$(z-z_0)(\bar{z}-\bar{z}_0) = \gamma^2$

$z\bar{z}-z_0\bar{z}-\bar{z}_0 z+z_0\bar{z}_0 = \gamma^2$

$z\bar{z}+(-z_0)\bar{z}+(-\bar{z}_0)z+(|z_0|^2-\gamma^2) = 0$ — (1)

i.e $\boxed{z\bar{z}+a\bar{z}+\bar{a}z+b=0}$ → $a=-z_0$, $b=|z_0|^2-\gamma^2$

i.e Centre $=-a$ & $\gamma = \sqrt{|a|^2-b}$

$$\text{i.e. } z\bar{z} + (-z_0)\bar{z} + (-\bar{z}_0)z + (|z_0|^2 - r^2) = 0 \quad \text{---(1)}$$

General eqn of Circle can also be given as;

Learn $$\boxed{z\bar{z} + a\bar{z} + \bar{a}z + b = 0} \quad \text{---(2)}$$

where Centre $= -a$ & rad $= \sqrt{|a|^2 - b}$

Proof Comparing (1) & (2), $a = -z_0$ & $b = |z_0|^2 - r^2$

i.e. $z_0 = -a$ & $r^2 = |z_0|^2 - b$

$\Rightarrow |z_0| = |a|$ & $r = \sqrt{|a|^2 - b}$

Q. the centre and radius of the circle?

$$z\bar{z} + (1-i)z + (1+i)\bar{z} - 7 = 0$$

$\bar{a} = 1-i$, $a = 1+i$, $b = -7$

on Comparision with (2)

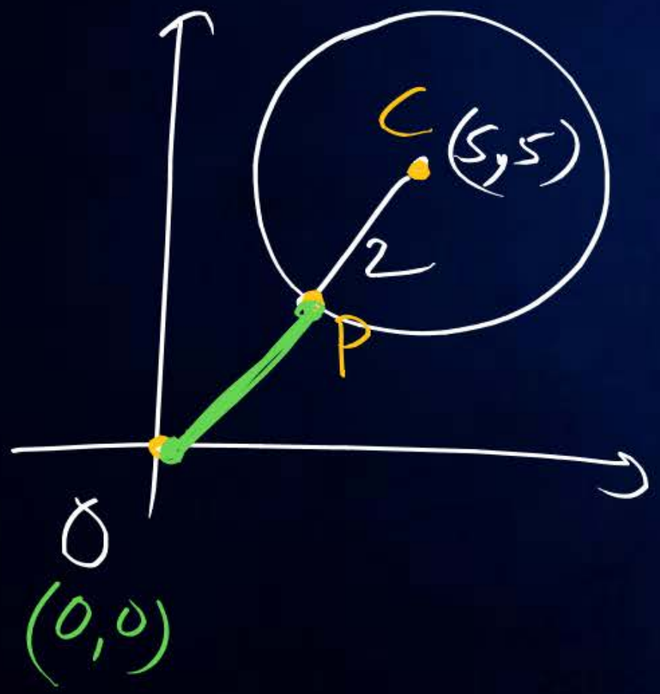
Centre $= -a = -(1+i) = -1-i \simeq (-1, -1)$

rad $= \sqrt{|a|^2 - b} = \sqrt{(\sqrt{2})^2 - (-7)}$

$= \sqrt{9} = 3$

Q: find the Min distance of any point on the curve $|z-5-5i|=2$ from origin

Sol: $|z-(5+5i)|=2$ (circle) $\Rightarrow$ Centre $= 5+5i \simeq (5,5)$, rad $= 2$



$OC = \sqrt{(5-0)^2+(5-0)^2} = \sqrt{50} = 5\sqrt{2}$

$PC = 2$ so $OP = OC - PC$

$= 5\sqrt{2} - 2$ Ans


Thank
YOU